

BRACT's

Vishwakarma Institute of Technology, Pune

Practical Implementation Sheet

Department: CSE-AI Semester: 5 Academic Year: 2026-27

Division: E Batch: 3 Practical No: 4

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Subject Name: Deep Learning

Problem Statement: Develop an LSTM-based model for time-series forecasting using stock price, weather, or sales datasets.

Algorithm:

- 1) Import the required libraries.
- 2) Load the stock price dataset.
- 3) Select the closing price values.
- 4) Normalize the data using Min-Max scaling.
- 5) Create sequences using the previous 60 days to predict the next day.
- 6) Split the data into training and testing sets.
- 7) Build and compile the LSTM model.
- 8) Train the model using the training data.
- 9) Predict the stock prices using the test data.
- 10) Calculate RMSE and MAE to evaluate the model.
- 11) Plot the actual and predicted stock prices.

Code:

```
!pip install yfinance -q

import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

import yfinance as yf

from sklearn.preprocessing import MinMaxScaler

from sklearn.metrics import mean_squared_error, mean_absolute_error
```

```
from tensorflow.keras.models import Sequential

from tensorflow.keras.layers import LSTM, Dense, Dropout


data = yf.download("AAPL", start="2015-01-01", end="2025-01-01")

data = data[['Close']]


scaler = MinMaxScaler(feature_range=(0, 1))

scaled_data = scaler.fit_transform(data)


sequence_length = 60

X = []

y = []


for i in range(sequence_length, len(scaled_data)):

    X.append(scaled_data[i-sequence_length:i, 0])

    y.append(scaled_data[i, 0])


X = np.array(X)

y = np.array(y)

X = X.reshape(X.shape[0], X.shape[1], 1)


split = int(0.8 * len(X))


X_train = X[:split]
```

```
X_test = X[split:]
```

```
y_train = y[:split]
```

```
y_test = y[split:]
```

```
model = Sequential([
```

```
    LSTM(50, return_sequences=True, input_shape=(X_train.shape[1], 1)),
```

```
    Dropout(0.2),
```

```
    LSTM(50),
```

```
    Dropout(0.2),
```

```
    Dense(25),
```

```
    Dense(1)
```

```
])
```

```
model.compile(optimizer='adam', loss='mean_squared_error')
```

```
model.fit(X_train, y_train, epochs=20, batch_size=32, validation_split=0.1)
```

```
predictions = model.predict(X_test)
```

```
predictions = scaler.inverse_transform(predictions)
```

```
actual = scaler.inverse_transform(y_test.reshape(-1, 1))
```

```
rmse = np.sqrt(mean_squared_error(actual, predictions))
```

```
mae = mean_absolute_error(actual, predictions)
```

```
print("RMSE:", rmse)

print("MAE:", mae)


plt.figure(figsize=(12, 6))

plt.plot(actual, label="Actual Price")

plt.plot(predictions, label="Predicted Price")

plt.title("LSTM Stock Price Forecasting")

plt.xlabel("Time")

plt.ylabel("Stock Price")

plt.legend()

plt.show()
```

Output:

```
Epoch 1/20
56/56 ————— 8s 62ms/step - loss: 0.0053 - val_loss: 8.3765e-04
Epoch 2/20
56/56 ————— 6s 71ms/step - loss: 9.4200e-04 - val_loss: 0.0010
Epoch 3/20
56/56 ————— 3s 54ms/step - loss: 8.4970e-04 - val_loss: 7.3732e-04
Epoch 4/20
56/56 ————— 3s 55ms/step - loss: 5.5954e-04 - val_loss: 0.0014
Epoch 5/20
56/56 ————— 7s 83ms/step - loss: 5.8236e-04 - val_loss: 6.7286e-04
Epoch 6/20
56/56 ————— 4s 54ms/step - loss: 4.7527e-04 - val_loss: 6.3155e-04
Epoch 7/20
56/56 ————— 3s 55ms/step - loss: 5.0400e-04 - val_loss: 5.9132e-04
Epoch 8/20
56/56 ————— 3s 60ms/step - loss: 5.8840e-04 - val_loss: 7.4843e-04
Epoch 9/20
56/56 ————— 4s 73ms/step - loss: 4.6027e-04 - val_loss: 6.2370e-04
Epoch 10/20
56/56 ————— 4s 54ms/step - loss: 4.4538e-04 - val_loss: 5.9667e-04
```

```

Epoch 11/20
56/56 ————— 3s 54ms/step - loss: 4.2029e-04 - val_loss: 5.4073e-04
Epoch 12/20
56/56 ————— 6s 72ms/step - loss: 4.2033e-04 - val_loss: 5.6639e-04
Epoch 13/20
56/56 ————— 3s 54ms/step - loss: 5.2177e-04 - val_loss: 8.2541e-04
Epoch 14/20
56/56 ————— 3s 54ms/step - loss: 4.3173e-04 - val_loss: 5.2265e-04
Epoch 15/20
56/56 ————— 7s 82ms/step - loss: 4.4964e-04 - val_loss: 5.7351e-04
Epoch 16/20
56/56 ————— 3s 55ms/step - loss: 4.4850e-04 - val_loss: 6.5512e-04
Epoch 17/20
56/56 ————— 3s 54ms/step - loss: 3.6118e-04 - val_loss: 5.6637e-04
Epoch 18/20
56/56 ————— 3s 54ms/step - loss: 4.2546e-04 - val_loss: 4.9440e-04
Epoch 19/20
56/56 ————— 5s 83ms/step - loss: 3.2817e-04 - val_loss: 5.5539e-04
Epoch 20/20
56/56 ————— 4s 65ms/step - loss: 3.9871e-04 - val_loss: 4.7982e-04
16/16 ————— 1s 34ms/step
RMSE: 6.6521218995626885
MAE: 5.307914454762528

```

